

Thomas Woodward

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Senior QA Automation Engineer

Quality Champion • AI Orchestration • Systems Validation

Senior QA Automation Engineer with 12+ years of experience building, maintaining, and scaling automation frameworks across complex software, cloud portals, and web applications. Experienced in **Ranorex (.Net/C#)**, **Playwright** and **Cypress (TypeScript)**, specialising in API test automation, custom fixtures, Page Object Models, and integrating regression packs into CI/CD pipelines.

I take a practical approach to engineering, leveraging AI-assisted development tools (Cursor, LLM coding assistants) to accelerate test creation and maintain suite reliability, **bypassing syntax bottlenecks to focus on test coverage, data accuracy, and robust API validation**. Focused on developer-led testing, exploratory testing, and working closely with cross-functional agile teams.

Areas of Expertise

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|------------------------------------|--------------------------------------|------------------------------|
| ✓ Quality Engineering | ✓ CI/CD Pipelines | ✓ Solutions Delivery |
| ✓ Shift-Left Automation Strategies | ✓ Microservice & Cloud Architectures | ✓ AI-Assisted Automation |
| ✓ Automation Engineering | ✓ Holistic systems Validation | ✓ Team Building & Leadership |

Quality Strategy & QA: Shift-Left Methodology, Test Pyramid Architecture, BDD (Gherkin), Exploratory Testing, Mobile Testing (iOS/Android/Kindle), Root Cause Analysis, Performance & Load Testing.

Automation & Frameworks: Cypress, Playwright, Vitest, Ranorex, Jest, API Testing (REST Client), Headless Validation.

Infrastructure & Orchestration: GitHub Actions (Matrix Workflows), Docker, Portainer, Proxmox, AWS, Release Ops, CI/CD Pipeline Design.

AI & Emerging Tech: LLM-Assisted Automation, AI Copilot Integration, Agent Orchestration (Task-to-Test), Prompt Engineering for QA.

Languages & Data: TypeScript, C#, Python, SQL, Zod (Schema Validation), MQTT, JSON, YAML, MATLAB.

Systems & Diagnostics: ADB, Xcode, Charles Proxy, Datadog, Home Assistant, Jira/Confluence Administration.

Compliance & Governance: 21 CFR Part 11, ISO Standards, COPPA, Storefront Certification.

Notable Achievements

- **Desktop & Hardware-in-the-Loop Automation (Ranorex):** Spearheaded modular, reusable Ranorex (C#/.NET) repository architectures and hardware-in-the-loop test frameworks for scientific desktop applications at Malvern Panalytical, ensuring seamless hardware-software validation under strict 21 CFR Part 11 and ISO compliance standards.
- **Playwright & API Test Automation:** Maintained production Playwright test suites at Synopsys/Ansys using custom API fixtures and Page Object Models for parallel CI execution. Published an open-source Playwright Automation showcase on [GitHub](#) demonstrating custom fixture architecture and API testing patterns.
- **AI-Assisted Test Acceleration:** Utilised AI tools (Cursor, Jira RoVo) to streamline test scenario generation and script maintenance, keeping regression suites in sync with rapid feature changes.
- **Developer-Aligned QE Frameworks:** Created modular test automation packages integrated into GitHub Actions CI/CD pipelines, enabling developers and QA team members to run fast, self-service regression checks.
- **Data & System Reliability:** Executed deep API mocking/stubbing, REST/GraphQL contract validation, and telemetry monitoring (Datadog) for high-performance cloud platforms, preventing downstream bugs in production.
- **Machine Learning Model Validation:** Validated deep learning models and data pipelines for Malvern Panalytical's NanoSight project using TensorFlow to automate nanoparticle tracking and ensure data accuracy.

Professional Experience

Synopsys / Ansys | Staff R&D Engineer | Feb 2023 – April 2026

Synopsys is a global leader in electronic design automation (EDA) and semiconductor IP, recently acquiring Ansys to expand into engineering simulation and physical AI.

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Designed and implemented test automation and quality engineering for the Ansys Cloud Portal. Built and maintained reusable, parallelised Cypress, Playwright, and Python automation suites integrated into CI/CD pipelines, collaborating closely with developers and product squads to ensure fast release feedback.

- **CI/CD Test Frameworks:** Created Cypress, Playwright and Python test scripts to validate REST/GraphQL APIs, backend data ingestion, and cloud compute workflows.
- **Automation Suite Maintenance:** Maintained core production test packs and conducted regular framework evaluations (comparing Playwright and Cypress) to keep execution fast and reduce test flakiness.
- **AI-Assisted Test Generation:** Used AI coding assistants (Cursor, Claude, Jira Rovo) to speed up test data creation and script writing, accelerating sprint delivery.
- **Observability & Triage:** Monitored Datadog telemetry to trace API execution health, identify root causes for failures, and set up release quality gates.
- **Collaborative QA:** Partnered with Stakeholders, Product Owners, and developers to review requirements early, refine acceptance criteria, and perform exploratory testing on complex sprint features.

Malvern Panalytical | Senior Verification & QA Engineer | Sept 2018 – Feb 2023

Malvern Panalytical creates scientific instrumentation and software for material characterisation in regulated industries, including Med Tech and Pharma.

Managed verification, compliance testing (ISO/21 CFR Part 11), and automated regression for complex scientific desktop and web applications.

- **Regulated Compliance:** Managed the testing process to ensure absolute compliance with strict ISO and 21 CFR Part 11 (FDA) standards for high-precision scientific data, while mentoring and guiding the team in best practices.
- **Ranorex & Hardware-in-the-Loop Automation:** Architected modular Ranorex (C#/.NET) automated regression suites and custom UI repositories, integrating hardware-in-the-loop validation for scientific instrumentation to accelerate release cycles.
- **Quality & Product Direction:** Acted as a core voice on quality across multiple projects (Mastersizer, Halo, Nanosight). I flagged critical issues on flagship products and helped launch the Nanosight line during a major hardware and software refresh.
- **AI Vision & Data Training:** Executed extensive data tagging and training for Nanosight's particle detection neural network, validating a computer vision model that eliminated manual tuning variance and improved baseline accuracy.
- **Manual & Automated Execution:** Defined and executed rigorous manual test scripts across desktop applications, whilst also managing automated suites (Ranorex, Selenium).
- **Root Cause & Risk-Based Testing:** Implemented risk-based approaches to make effective use of resources, driving quality defect logging, triage, and root cause analysis.
- **Quality Advocacy:** Co-founded the global UX team and championed a human-centred design culture, pushing for quality and usability across the entire business.

SEGA Hardlight | QA Engineer | Aug 2017 – May 2018

- **High-Pressure Delivery:** Managed COPPA compliance, feature sign-off and submission testing for multiple global Top 10 hits including *Sonic Dash*, *Sonic Boom* & *Sonic Forces: Speed Battle*, remaining steady under pressure to make sound, risk-based decisions when requirements shifted.

Connect Group | QA & Automation Engineer | Oct 2016 – Aug 2017

- **Cross-functional Ownership:** Led test automation and manual testing across multiple teams including front end, core platform, and live operations.
- **Financial & Logic Integrity:** Developed Java-based automation suites to validate complex calculators and REST APIs, focusing on mathematical accuracy and backend integrity.

Adactus / EDM Group | Junior QA Tester | Oct 2015 – Jul 2016

- **Data Integrity:** Validated monthly SQL reports for hospitality stakeholders, ensuring reporting integrity and data accuracy for critical systems.
- **Mobile Automation:** Executed manual test scenarios and mobile automation using Java.

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Education & Qualifications

MEng Computer Games Design & 3D Modelling • Staffordshire University

Computer Science & Applied IT • Stourbridge College